

Sidharth Sreeram

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University of Waterloo Mechatronics Engineering graduate with hands-on experience designing and developing robust software systems across a range of applications. Proficient in Python, C++ and modern development frameworks, with a strong foundation in problem-solving, system design, and data processing.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, SQL, Bash

Frameworks & Libraries: PyTorch, NumPy, Matplotlib, OpenCV, Pandas, Flask

Development: Linux, Git, LaTeX, Markdown, JSON, VSCode, Docker, GitHub, Azure DevOps, Jira, Agile

EDUCATION

University of Waterloo | *B.ASc in Mechatronics Engineering*

June 2026

– **Relevant Courses:** Cooperative and Adaptive Algorithms, Intro to Machine Learning

EXPERIENCE

Research Assistant - UW WISE Lab | *Python, PyTorch*

Jan – Apr 2026

- Researched and developed **Python** logic to organize visual and text data into structured VLM prompt format.
- Aggregated NuScenes-QA, NuCaptions, and NuX datasets with multi-sensor metadata by sample token.
- Reconstructed the **OpenDriveVLA** Stage-2 prompt spec, fusing camera, LiDAR, and CAN-bus ego state.
- Generated **728K** JSONL instruction-tuning samples for camera-vs-LiDAR VLA reliability research.

Software Engineering Analyst - Tamaki Controls | *Python, Java, SQL*

May – Aug 2025

- Designed an **API** response schema improving cross-service communication between **Java** and **Python**.
- Built 6 customer-facing **REST** endpoints importing and exporting UTF-8 encoded CSV/JSON data.
- Wrote **SQL** queries aggregating source data from the database backing each export endpoint.
- Updated **API documentation** with input/output descriptions and example Python client scripts.

Platform Software Engineer - Ford Motor Company | *Python, C++, GCov*

Sept – Dec 2024

- Developed a **Python CLI** for the AUTOSAR State Management API in a containerized Linux environment.
- Automated 6-stage state transitions, replacing stage-by-stage manual navigation with one command.
- Drove CLI adoption across **3 engineering teams** for state control in testing and integration.
- Created C++ test suites with GCov, raising line coverage **45% to 94%** and condition coverage **50% to 85%**.

Controls System Engineer - Brock Solutions | *Python, SQL*

Jan – Apr 2024

- Built SCADA HMI screens visualizing live coil crane, acid bath, and pump operations at a steel facility.
- Simulated PLC outputs across a **27-PLC** system to validate HMI designs pre-commissioning.
- Wrote Python scripts generating synthetic time-series data to demo plant behavior to customers.

PROJECTS

Qwen 3 VQA Model Tuning | *Python, PyTorch, Qwen 3, VLM, LoRA, Hugging Face*

- Fine-tuned Qwen 3 to perform driving-scene QA reasoning using camera and LiDAR data from the NuScenes dataset.
- Built a training pipeline using LoRA, quantization, and HuggingFace TRL to perform under limited GPU resources.

UWB Tracking System | *Typescript, REST Api, PostgreSQL, Redis*

- Built a distributed backend: a REST API with PostgreSQL and a Redis cache serving a React frontend.
- Designed services to ingest streaming timestamps, compute positions every second, and serve results.
- Achieved **~30cm** accuracy via TDoA Localization with the FireFly algorithm and weighted least squares optimization
- Collaborated with a 5-member team to define project requirements, and tracked progress through bi-weekly meetings.

Path Following Robot | *Python, Raspberry Pi, OpenCV, Git*

- Designed and implemented a real-time computer vision system for color-based object detection and path following.
- Applied a camera homography transformation, enabling accurate navigation and target localization.